Revisiting Fairness Evaluation under Simpson's Paradox

A Stratified Analysis of Link Prediction Fairness

Weixiang Wang and Sucheta Soundarajan
Syracuse University

The 6th ISCS, June 03, 2026, La Rochelle, France

Code available at: github.com/Oakwoo/stratified-disparity

Why Fairness in Link Prediction Matters

Link prediction algorithms are widely used in many real-world systems.

- Friend Recommendation
- Collaboration Prediction
- Citation Recommendation
- Social Exposure

Link prediction algorithms influence visibility and opportunities.

- Performance disparities across groups may reinforce existing inequalities.

Existing fairness metrics evaluate only aggregate group performance.

Why Aggregation Fails in Networks

Groups Occupy Different Structural Regions

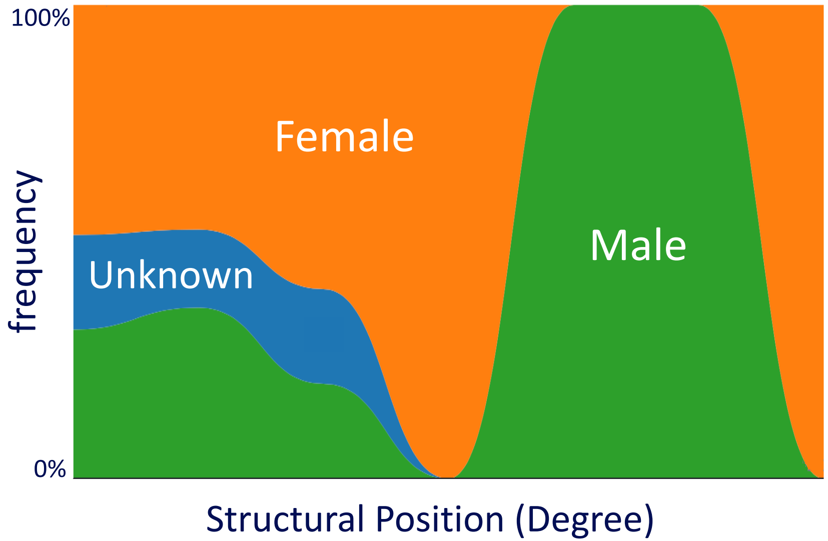


Illustration based on the DBLP Graphics co-authorship network

Structural heterogeneity is common in real world networks.

Power-law topology amplifies this heterogeneity.

- high-degree hubs
- sparse peripheral nodes

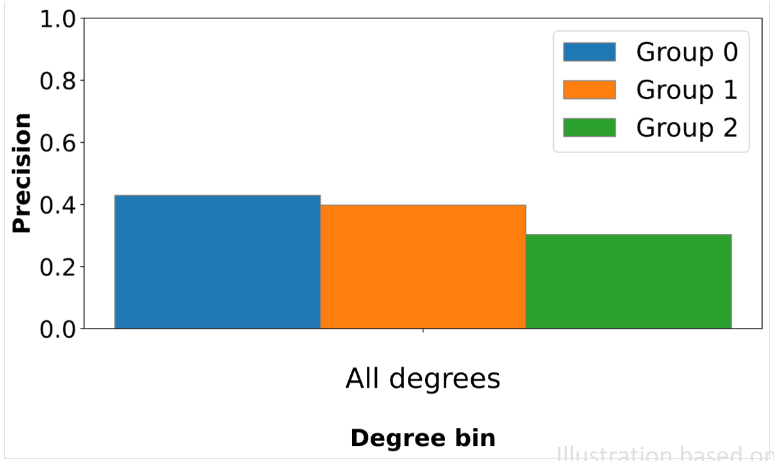
Aggregation may hide or amplify subgroup-level disparities.

This creates Simpson's Paradox in fairness evaluation.

Simpson's Paradox in Link Prediction Fairness

Aggregate disparity nearly disappears after degree stratification: 0.0657 → 0.0099

Aggregate Comparison



Degree-Stratified Comparison

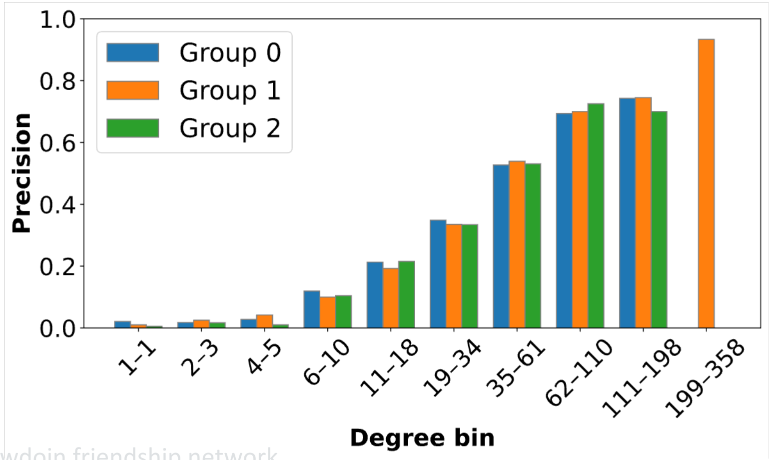


Illustration based on the Facebook Bowdoin friendship network

Aggregate disparity largely reflected **structural imbalance** rather than **within-group performance differences.**

Definition of Stratified Disparity

Fairness is evaluated as expected disparity within comparable structural contexts.

$$StratifiedDisparity(n) = \sum_{j=1}^{n} \mathrm{Prob}_j \times \mathrm{Diff}_j(g_1, \ldots, g_t)$$

$g_1, \ldots, g_t$ is group performance within bin j.

Diff_j is disparity across protected groups.

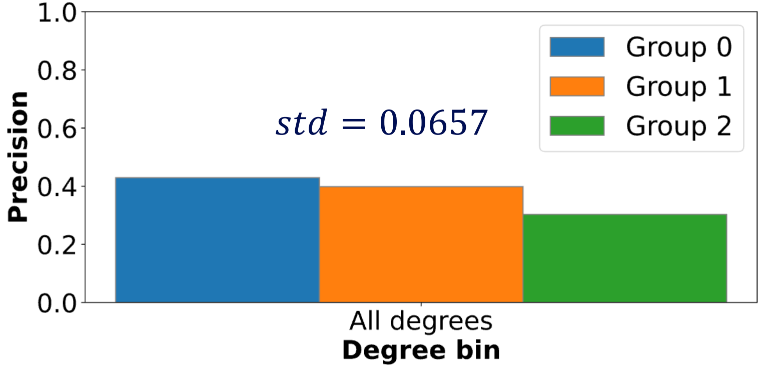
Prob_j is bin weight

Aggregate within-bin disparities across structural bins.

$n = 1$ corresponds to traditional aggregate fairness evaluation.

Computing Stratified Disparity

Aggregate Disparity (n=1)



Degree-Stratified Disparity (n=3)

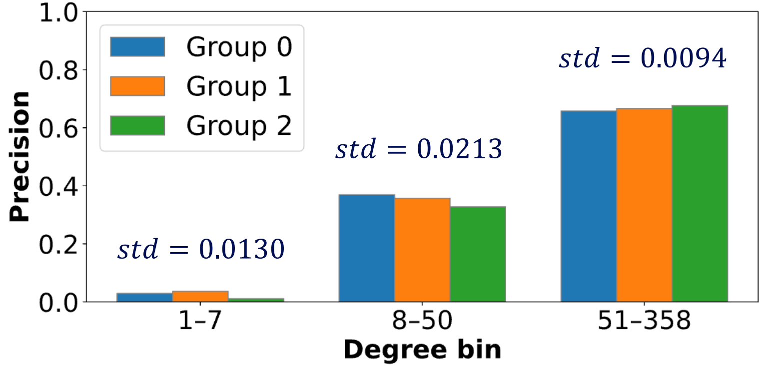


Illustration based on the Facebook Bowdoin friendship network

$$StratifiedDisparity(1) = 100\% \times 0.0657 = 0.0657$$

$$StratifiedDisparity(3) = 10\% \times 0.0130 + 64.6\% \times 0.0213 + 25.4\% \times 0.0094 = 0.0174$$

Bin weights determine how much each local disparity contributes.

Stratified Disparity Curve and Elbow Point

Stratified disparity changes as the number of bins increases

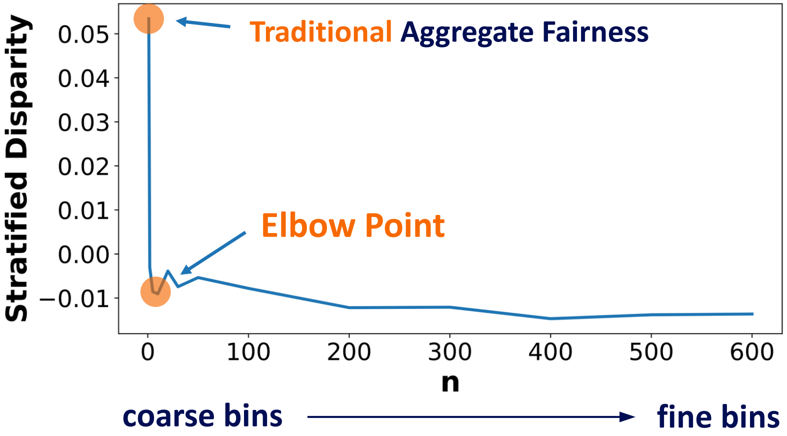


Illustration based on the Facebook Bowdoin friendship network

Too few bins: hidden structural heterogeneity

Too many bins: sparse and noisy estimates

Elbow point: captures structural patterns without becoming too noisy.

The elbow point identifies the most informative level of structural stratification.

Experimental Setup

Extensive evaluation across 8 datasets and 8 link prediction methods

Datasets

Social Networks:

Dutch, Bowdoin, UNC, Pokec

Collaboration Networks:

DBLP Datamining, DBLP Graphics

Others:

Polblogs, Recidivism

Methods

Accuracy-oriented:

Jaccard, DeepWalk

Fairness-aware pre-processing:

FairLP, UGE, EDITS

Fairness-aware embeddings:

FairWalk, CrossWalk, FLIP

Curve Insights

n=1: Aggregate disparity

Elbow point

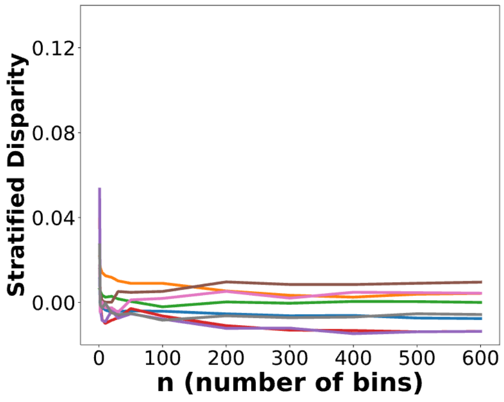
↓ Trend: Structure explains bias

↑ Trend: Hidden bias revealed

Analyze how structural heterogeneity shapes fairness evaluation.

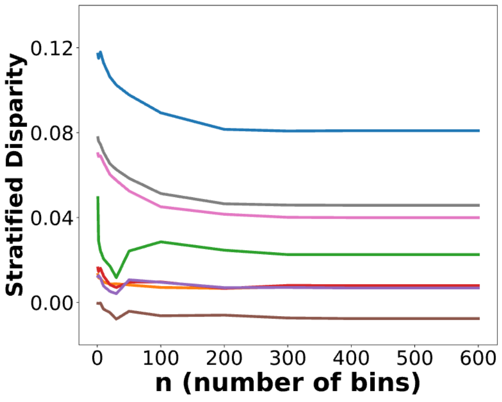
Three Fairness Behaviors Revealed by Stratification

Topology-Induced Unfairness



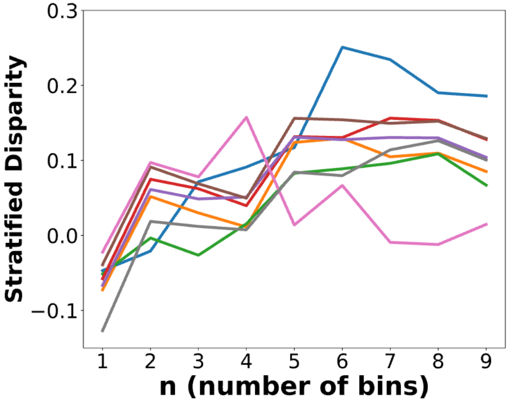
Aggregate disparity **emerges** from structural imbalance.

Power-Law Masked Unfairness



Disparity **persists** even after structural conditioning.

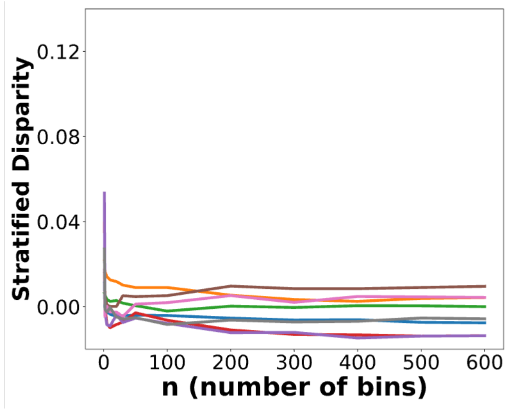
Hidden Subgroup Unfairness



Aggregate fairness **masks** subgroup-level disparities.

Three Fairness Behaviors Revealed by Stratification

Topology-Induced Unfairness



Aggregate disparity **emerges** from structural imbalance.

Power-Law Masked Unfairness

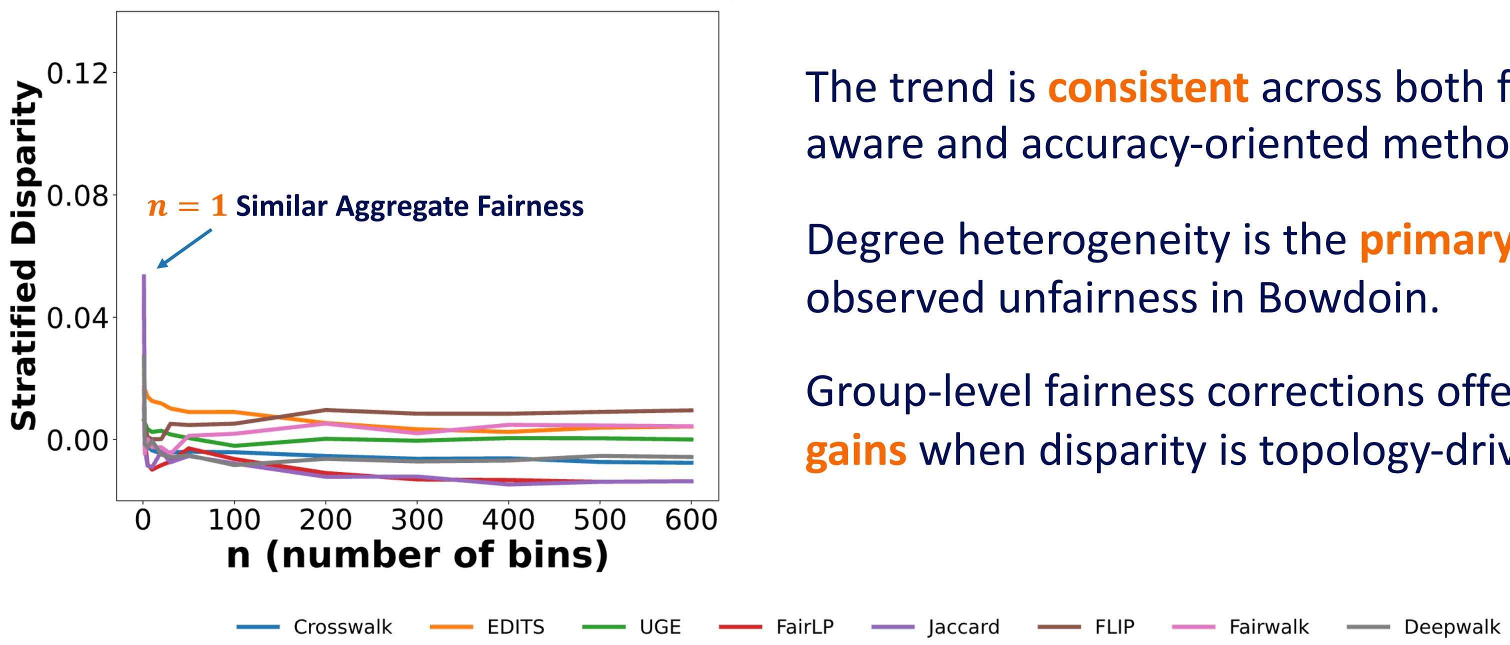
Disparity **persists** even after structural conditioning.

Hidden Subgroup Unfairness

Aggregate fairness **masks** subgroup-level disparities.

Topology-Induced Unfairness in Bowdoin

Aggregate disparity rapidly disappears after controlling for degree.



The trend is **consistent** across both fairness-aware and accuracy-oriented methods.

Degree heterogeneity is the **primary driver** of observed unfairness in Bowdoin.

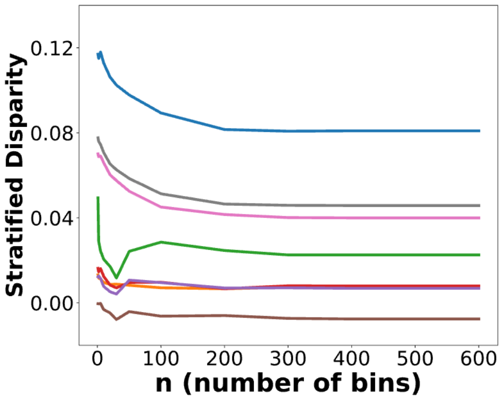
Group-level fairness corrections offer **limited gains** when disparity is topology-driven.

Three Fairness Behaviors Revealed by Stratification

Topology-Induced Unfairness

Aggregate disparity **emerges** from structural imbalance.

Power-Law Masked Unfairness



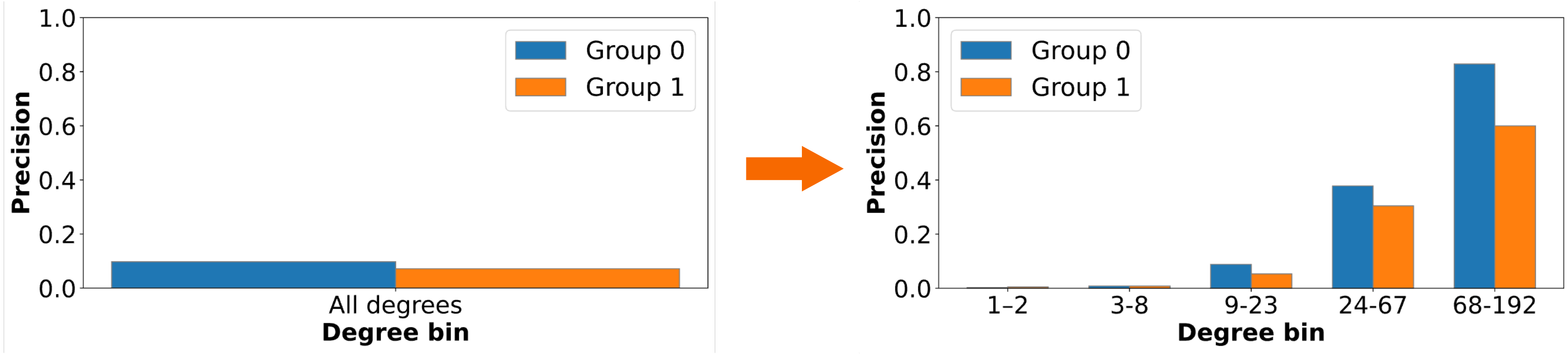
Disparity **persists** even after structural conditioning.

Hidden Subgroup Unfairness

Aggregate fairness **masks** subgroup-level disparities.

Power-Law Masked Unfairness in PolBlogs

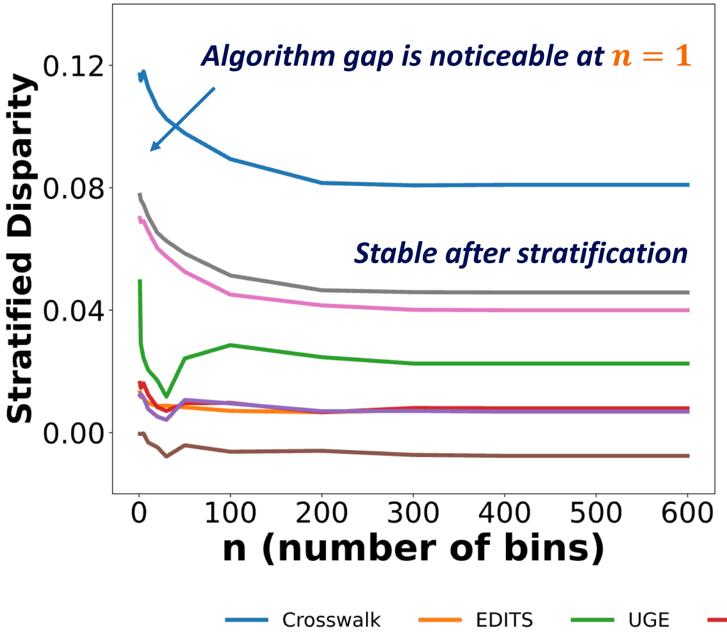
Large subgroup disparities remain hidden under aggregate evaluation.



Power-law degree distribution gives low-degree bins most of the weight, masking high-degree disparities.

Algorithm Choice Matters in PolBlogs

Stratified disparity remains relatively stable across bin counts.



Degree stratification does not explain away the disparity.

Algorithm design becomes critical.

- FLIP performs best
- CrossWalk performs worse than DeepWalk

In PolBlogs, suppressing attribute signals works better than amplifying cross-group walks.

Three Fairness Behaviors Revealed by Stratification

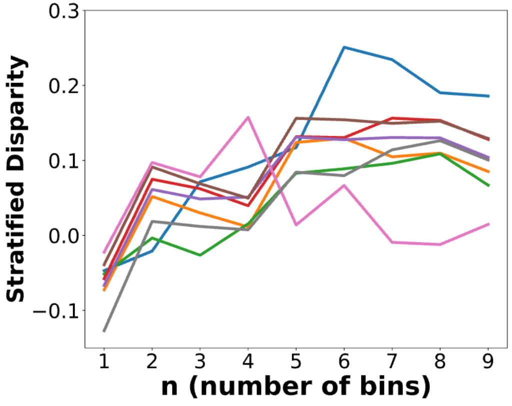
Topology-Induced Unfairness

Aggregate disparity **emerges** from structural imbalance.

Power-Law Masked Unfairness

Disparity **persists** even after structural conditioning.

Hidden Subgroup Unfairness



Aggregate fairness **masks** subgroup-level disparities.

Hidden Unfairness Through Cancellation

Opposing subgroup advantages cancel out under aggregate evaluation.

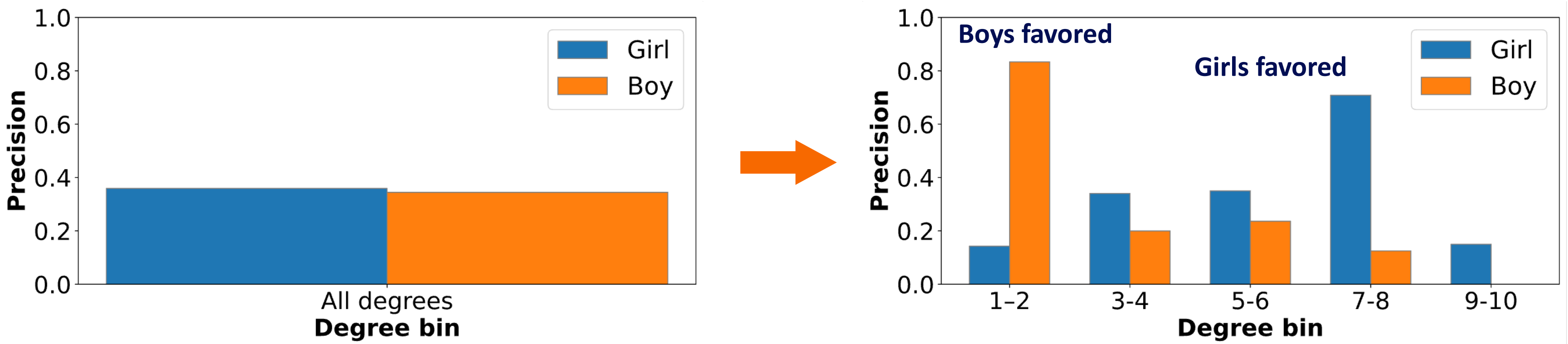


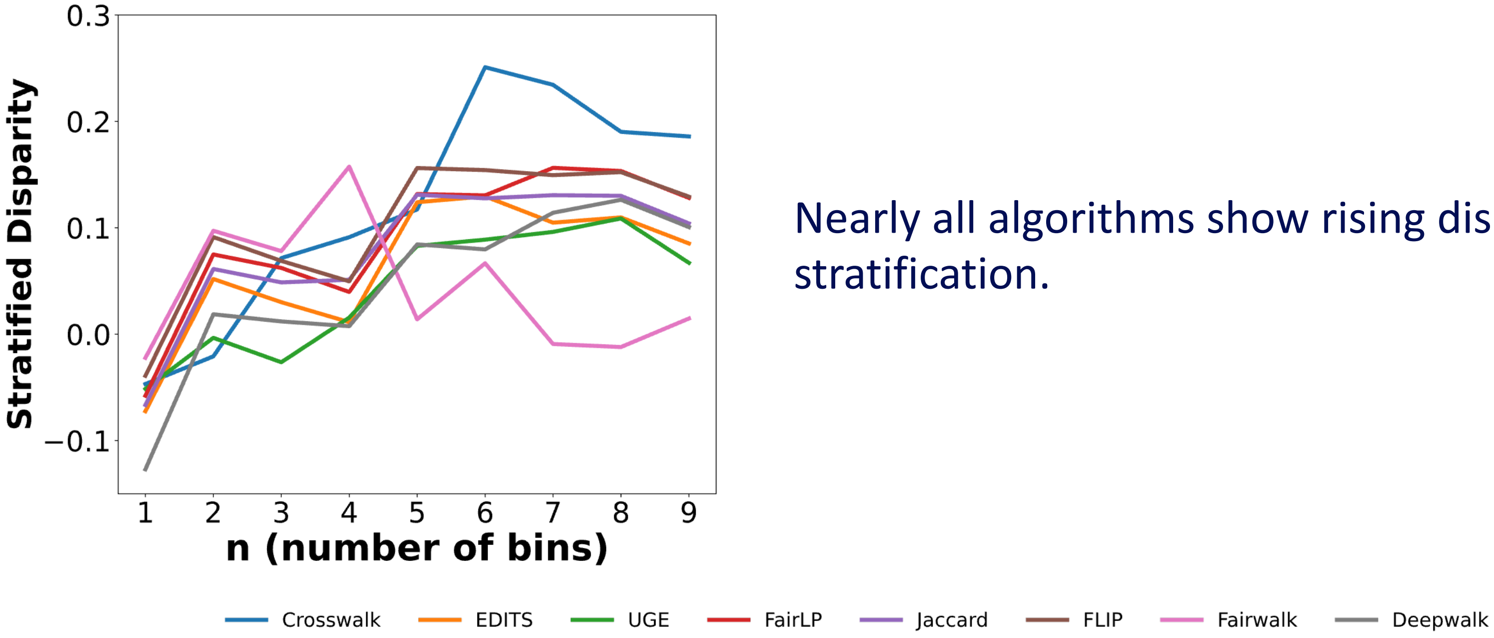
Illustration based on the Dutch School Network

Aggregate precision looks similar across groups.

Stratification reveals opposing subgroup advantages that cancel out in aggregate.

Hidden Subgroup Unfairness in Dutch School

Stratification reveals disparities hidden by aggregate evaluation.



Nearly all algorithms show rising disparity after stratification.

Takeaways

Aggregate fairness can misidentify the source of unfairness.

Traditional group-level evaluation may confuse protected-attribute effects with structural imbalance.

Stratification reveals how topology shapes fairness.

Stratified Disparity separates aggregate disparity, hidden subgroup disparity, and power-law masked disparity.

Fairness mitigation should depend on the diagnosed mechanism.

When disparity is topology-driven, group-level correction may help little; when it is attribute-driven, algorithm design becomes critical.

Thank you!

Paper

Scan for paper PDF

Code

Scan for GitHub Repo